

4.1

$$z = ax + by + c$$

$$(0, 0, 1), (1, 0, 1), (0, 1, 1), (1, 1, 2), (1, -1, 0), (-1, 1, 1)$$

$$(x_i, y_i, z_i)$$

primi slovese pro x_i , drugi y_i a tretji $c = 1 \cdot c$

~~$$A = \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix}$$~~

$$A = \begin{pmatrix} 0 & 0 & 1 & 1 \\ 1 & 0 & 1 & 1 \\ 0 & 1 & 1 & 1 \\ 1 & 1 & 1 & 2 \\ 1 & -1 & 1 & 0 \\ -1 & 1 & 1 & 1 \end{pmatrix}$$

$$A^T A x = A^T b$$

$$A^T A = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 0 & 0 & 1 \\ 1 & 0 & 1 \\ 0 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & -1 & 1 \\ -1 & 1 & 1 \end{pmatrix} = \begin{pmatrix} 4 & -1 & 2 \\ -1 & 4 & 2 \\ 2 & 2 & 6 \end{pmatrix}$$

$$A^T b = \begin{pmatrix} 0 & 1 & 0 & 1 & 1 & -1 \\ 0 & 0 & 1 & 1 & -1 & 1 \\ 1 & 1 & 1 & 1 & 1 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 2 \\ 4 \\ 6 \end{pmatrix}$$

$$A x = b \sim \begin{pmatrix} 4 & -1 & 2 & 2 \\ -1 & 4 & 2 & 4 \\ 2 & 2 & 6 & 6 \end{pmatrix} \sim \begin{pmatrix} -1 & 4 & 2 & 4 \\ 4 & -1 & 2 & 2 \\ 2 & 2 & 6 & 6 \end{pmatrix} \sim \begin{pmatrix} 1 & -4 & -2 & -4 \\ 4 & -1 & 2 & 2 \\ 2 & 2 & 6 & 6 \end{pmatrix} \sim$$

$$\begin{pmatrix} 1 & -4 & -2 & -4 \\ 0 & 15 & 10 & 18 \\ 0 & 10 & 10 & 14 \end{pmatrix} \sim \begin{pmatrix} 1 & -4 & -2 & -4 \\ 0 & 15 & 10 & 18 \\ 0 & 0 & 10 & 6 \end{pmatrix}$$

$$c = 0,6 \quad y = 0,8 \quad x = 0,4$$

$$z = 0,4x + 0,8y + 0,6$$